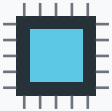
Computer Architecture • Chapter 1

General Organization of the Central Unit



history, main parts, Von Neumann and Harvard, memory, buses

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Course: UEF31, Architecture des ordinateurs

Goals of this chapter

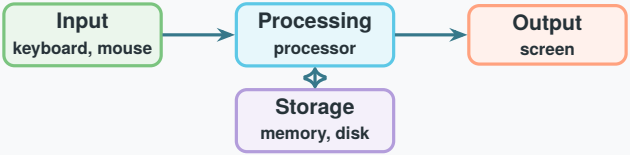
At the end, you can:

- name the main parts of a computer and their jobs;
- explain the Von Neumann and Harvard models;
- compute a memory size, and describe the three buses.

What is a computer?

Definition

A machine that takes **data**, processes it with a **program**, and gives **results**.



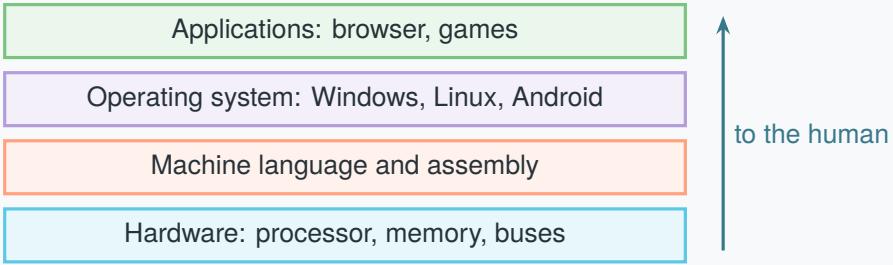
Analogy

A kitchen: data = ingredients, program = recipe, processor = cook.

Example: computing 25×4

1. **Input**: you type 25, $\times$, 4: each key becomes a number code.
2. **Storage**: the two numbers are kept in memory.
3. **Processing**: the processor runs the multiply instruction: 100.
4. **Output**: the result goes to the screen.

The layers of a computer

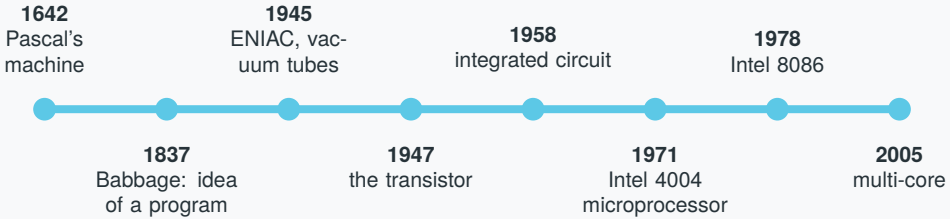


This course works on the **two lowest layers**: hardware and assembly.

How does a program run?

1. The program is a file on the **disk** (SSD or HDD).
2. The system **copies it into RAM**: the disk is too slow for the processor.
3. The processor reads the instructions **one after the other** and runs them.

A short history



Moore's law (1965)

The number of transistors on a chip **doubles about every two years**.

The four generations

Generation	Period	Technology	Example
1st	1945–1955	vacuum tubes	ENIAC (30 tons)
2nd	1955–1965	transistor	IBM 7090
3rd	1965–1980	integrated circuits	IBM 360
4th	1980–today	microprocessor	PC, phone

Each generation made the parts **smaller**, **faster** and **less hungry for energy**.

How does a computer count?

Units

- **bit**: 0 or 1 **byte**: 8 bits **word**: what the processor handles at once (16 bits in the 8086).

Dec	Bin	Hex	Dec	Bin	Hex
5	0101	5	10	1010	A
9	1001	9	12	1100	C
			15	1111	F

Converting numbers

$2F_H$ to binary and decimal

$2 = 0010$, $F = 1111$, so $2F_H = 0010\,1111_2$ and $2 \times 16 + 15 = 47$.

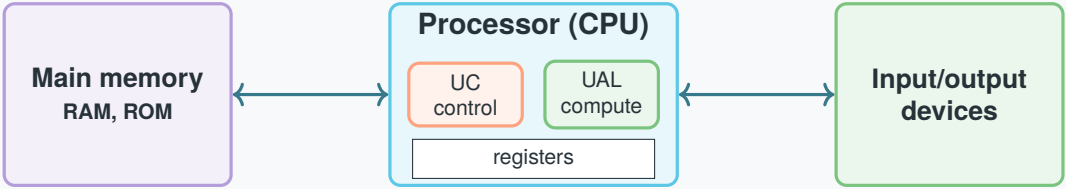
1011_2 to decimal

Weights 8, 4, 2, 1: $8 + 0 + 2 + 1 = 11$.

13 to binary: divide by 2

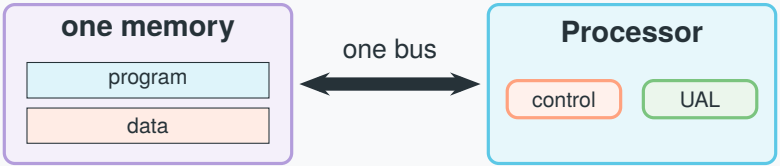
Remainders 1, 0, 1, 1 read backwards: 1101_2.

The three main parts



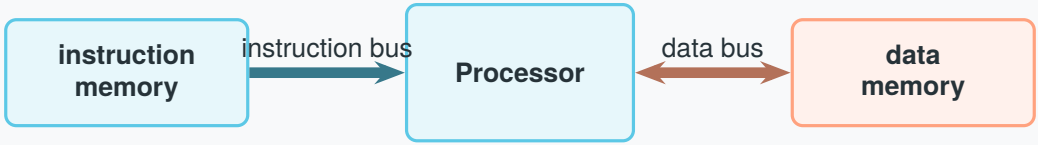
The **central unit** = the processor + the main memory.

The Von Neumann model (1945)



- **Stored program**: instructions and data in **the same memory**.
- Drawback: one bus, so one transfer at a time: the **Von Neumann bottleneck**.

The Harvard model



Analogy

A library with **two doors** instead of one: two queues move at the same time.

Von Neumann or Harvard?

	Von Neumann	Harvard
Memory	one, shared	two, separate
Speed	slower (bottleneck)	faster
Cost	simple, cheap	more complex
Used in	PCs, servers	microcontrollers, DSP

Modern PCs mix both: shared RAM, but separate L1 caches for code and data (**modified Harvard**).

The processor and its clock

Part	Job	Analogy
control unit (UC)	reads and orders	orchestra conductor
UAL	computes	calculator
registers	tiny fast memory	scrap paper

A 3 GHz clock

$T = \frac{1}{3 \times 10^9} \approx 0.33$ ns per cycle: 300 million cycles while you blink.

Memory: boxes with addresses

address	content
0000	3A
0001	FF
0002	07
0003	41
0004	00

- **Read**: send an address, get the content.
- **Write**: the new value replaces the old one.

Analogy

Address = house number, content = who lives there.

Types of memory

Type	Description	Power off	Used for
SRAM	transistors, very fast	erased	cache
DRAM	capacitors, refreshed	erased	main memory
ROM	written in the factory	kept	old boot programs
EEPROM / Flash	erased electrically	kept	BIOS/UEFI, SSD, USB

Volatile memory loses its content without power (RAM); **non-volatile** memory keeps it.

Computing a memory size

Rule

n address lines $\Rightarrow 2^n$ addresses. Size $= 2^n \times$ (size of one box).

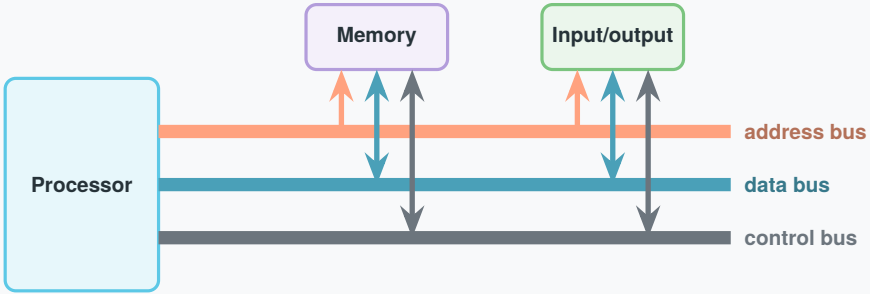
The 8086: 20 address lines

$2^{20} = 1024 \times 1024 = 1\,048\,576$ bytes $= 1$ MB.

How many lines for 64 KB?

$64 \times 1024 = 2^6 \times 2^{10} = 2^{16}$, so 16 lines.

The three buses



Address: one way (from the processor). Data and control: both ways.

Example: the processor reads box 0003

1. It puts 0003 on the **address bus**.
2. It sends **read** (RD) on the **control bus**.
3. The memory puts 41_H on the **data bus**, and the processor keeps it.

Analogy

The post office: the shelf number, the parcel, and the words “give me”.

Bus width

- Address bus width $\Rightarrow$ **maximum memory**.

- Data bus width $\Rightarrow$ **bytes moved at once**: 1000 bytes take 1000, 500 or 125 transfers on 8, 16 or 64 bits.

Processor	Data bus	Address bus	Max memory
8086	16	20	1 MB
80386	32	32	4 GB
x86-64	64	48	256 TB

Practice

Try it

1. Convert $3C_H$ to decimal. — 60
2. Max memory with a 24-bit address bus? — 16 MB
3. Instructions per second at 2 GHz with 4 cycles each? — 500 million

Summary

Remember

- A computer = processor + memory + input/output, linked by buses.
- Von Neumann: one memory; Harvard: two memories.
- n address lines give 2^n addresses.